

Dhruv Gupta

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EDUCATION

Master of Science, Data Science – University of Colorado Boulder, USA: GPA: 3.9/4 May 2025
Machine Learning (ML), Statistical Methods, Natural Language Processing, Data Mining, Information Visualization, Ethics in Data Science.

Bachelor of Technology – Amity University, INDIA: May 2019
Mathematics (Linear Algebra, Calculus, Discrete Mathematics, Statistics), Data Structures and Algorithms, Database Management Systems.

SKILLS

Languages & Libraries: Python (Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow), PostgreSQL, SQL, NoSQL, R, Bash, C++

Data Infrastructure (AWS): S3, Glue, Lambda, Athena, Lake Formation, EC2, CloudWatch, EventBridge, QuickSight, Kinesis

Big Data & Distributed Compute: Apache Spark, PySpark, Hadoop, MapReduce, Databricks, Kubernetes

Streaming & Workflow Orchestration: Apache Airflow, event-driven pipelines (EventBridge/SNS), Databases, Data Warehousing

Machine Learning: NLP, Statistical Analysis, Regression, Classification, Feature Engineering, Hypothesis Testing, Web & API Scraping

Data Engineering: Distributed ETL/ELT, Data Modeling, Data Quality, Data Governance, Data Lineage

Tools: Snowflake, BigQuery, IBM DB2, SQL Server, Git, GitHub, Jira, Tableau, Power BI, Informatica, Talend

Certifications: Databricks Certified Data Engineering Associate, Microsoft Azure AI Fundamentals AI 900.

EXPERIENCE

Data Engineer – VP Health Systems Jan 2025 – Present

- Designed and maintained distributed AWS-based data infrastructure using S3, Glue, Lambda, Athena, and event-driven patterns to support high-scale ingestion and transformation workflows.
- Built automated Python + SQL data pipelines to process structured and semi-structured datasets with strong controls around data quality, validation, logging, and monitoring.
- Established robust data integrity checks using CloudWatch logs, Glue validations, and schema consistency rules to ensure accurate metrics across the organization.
- Partnered with engineering and product teams to define data consumption patterns, access models, and scalable transformation logic.
- Designed and documented pipeline architectures, data flows, and operational procedures to support reliable analytics.

Data Science Research Assistant – Business Research Division, University of Colorado Boulder Jul 2024 – Oct 2025

- Developed Python-based workflows to clean, transform, validate, and publish large analytical datasets.
- Automated ingestion processes, reducing refresh times by **20%**, and strengthened reproducibility through careful documentation.
- Built datasets powering analytical dashboards and supported downstream researchers with well-structured consumption patterns.

Data Engineering Analyst – Accenture Sept 2019 – Jul 2023

- Designed and automated big-data pipelines using Python, Apache Beam, Airflow, and GCP for enterprise-scale ingestion and transformation.
- Unified data from Snowflake, IBM DB2, SQL Server, and BigQuery into analytical warehouses used for enterprise reporting.
- Optimized large-scale SQL workloads using UDFs, CTEs, and warehouse modeling, improving performance by 50%.
- Ensured system-wide data accuracy through pipeline debugging, lineage documentation, and root-cause analysis across distributed systems.
- Collaborated with engineers, BI teams, and business users to define requirements and deliver reliable, scalable data infrastructure solutions.

PROJECTS

AWS Data Lake Creation: PostgreSQL Server, S3, Lake Formation, Glue, Crawler, Athena

- Designed Glue ETL pipelines to migrate PostgreSQL data to S3 and convert raw data to optimized Parquet, improving query performance by 40%.
- Implemented automated metadata discovery and governance to support consistent downstream consumption.

Draft Order Automation (Ongoing) - SNS, S3, Lambda, Glue, Gemini Flash 2.5, EventBridge

- Designing an event-driven ingestion and transformation pipeline for unstructured retailer order emails, converting them into automated draft purchase orders.
- Implementing monitoring, error handling, and scalable orchestration patterns.

ACHIEVEMENTS

Accenture Pinnacle Award: Rising Star Category- FY 22 Q1 Mar 2022

Awarded pinnacle award for successful completion of SAP BO to Power BI dashboard migration project.

Accenture Celebrates Excellence (ACE) Award- FY21 Q3 Aug 2021

Received recognition for automating error resolution process saving 50 hours per month per team.